

EXACT FINITE-TIME SYNCHRONIZATION FOR RANDOM CAP–FLOOR INTERVAL COCYCLES

ANONYMOUS

ABSTRACT. Independently at each time, choose a threshold $U \in [0, 1]$ from an atomless law and apply either the floor map $x \mapsto \max\{x, U\}$ with probability p or the cap map $x \mapsto \min\{x, U\}$ with probability $1 - p$. We give an exact finite-word normal form: every composition is either an interval clamp or a constant, and it becomes constant precisely when the largest floor threshold crosses the smallest cap threshold. If T is the first constant time, then for every atomless threshold law

$$\mathbb{P}(T > n) = \sum_{j=0}^n p^j (1-p)^{n-j}.$$

For $0 < p < 1$, the probability generating function identifies T with the sum of independent geometric variables of parameters p and $1 - p$; in particular its mean and variance are explicit. For uniform thresholds, the expected image diameter is exactly $\mathbb{P}(T > n)/(n + 1)$. This yields an annealed exponential rate, a critical cancellation at $p = 1/2$, and polynomial endpoint decay at $p \in \{0, 1\}$. Almost every mixed path instead has diameter exactly zero after a finite time, so its logarithmic behavior is finite absorption rather than a conventional finite Lyapunov exponent. The note is owner-subtracted against general theories of iterated random functions, monotone synchronization, and finite-chain contraction semigroups, and makes no absolute novelty or priority claim.

1. INTRODUCTION

Let a random map of the unit interval either push every point up to a fresh threshold or pull every point down to it. Each generator is monotone and idempotent, but alternating the two types creates an irreversible event: once a lower threshold reaches an upper threshold, the entire interval has coalesced to a single point. The event is governed by ranks rather than by the numerical threshold distribution.

General iterated-random-function methods, including contractive-on-average criteria and invariant-law constructions, have a broad established theory [1]. Synchronization for iid monotone random maps on Euclidean spaces is also treated in a substantially more general setting [2]. On a finite chain, semigroups generated by contraction mappings have their own structural literature [3]. Those frameworks, and the general phenomenon of monotone random synchronization, are not claims of this note.

The residual calculation here is deliberately narrow and exact. We combine a finite-word clamp normal form with exchangeability of iid ranks. This produces the complete survival law, a geometric-sum representation of the synchronization time, its moments, and an exact uniform-threshold diameter formula. It also separates two meanings of contraction that can otherwise be conflated: pathwise finite absorption and exponential decay of a positive finite-time expectation.

External release is **HOLD**. The proofs and exact controls support the displayed conjunction for this particular cap–floor family, but no claim is made that an equivalent specialization is absent from the iterated-function, synchronization, order-statistic, or semigroup literatures.

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2. MODEL AND FINITE-WORD NORMAL FORM

Put $I = [0, 1]$. For $u \in I$, define the floor and cap maps

$$(1) \quad F_u(x) = \max\{x, u\}, \quad C_u(x) = \min\{x, u\}.$$

For $0 \leq a < b \leq 1$, write

$$(2) \quad K_{a,b}(x) = \max\{a, \min\{x, b\}\},$$

and let κ_c denote the constant map with value c .

The order of composition matters. Given generators $g_1, \dots, g_n$, our forward convention is

$$(3) \quad \Phi_0 = \text{Id}, \quad \Phi_n = g_n \circ g_{n-1} \circ \dots \circ g_1.$$

Thus g_1 acts first and g_n acts last. Define the two running threshold records

$$(4) \quad A_n = \max(\{u_i : 1 \leq i \leq n, g_i = F_{u_i}\} \cup \{0\}), \quad B_n = \min(\{u_i : 1 \leq i \leq n, g_i = C_{u_i}\} \cup \{1\}).$$

For a concrete order check,

$$(5) \quad C_{1/4} \circ F_{3/4} = \kappa_{1/4}, \quad F_{3/4} \circ C_{1/4} = \kappa_{3/4}.$$

Thus reversing the word can change the collapsed value even though both words have the same two threshold records.

Theorem 2.1 (Clamp-or-constant normal form). *For every finite word in the generators (1), exactly one of the following alternatives holds.*

(1) *If $A_n < B_n$, then*

$$\Phi_n = K_{A_n, B_n}, \quad \Phi_n(I) = [A_n, B_n].$$

(2) *If $A_n \geq B_n$, then Φ_n is a constant map.*

The value of the constant in the second alternative generally depends on chronological order, not only on (A_n, B_n) .

Proof. Suppose first that the current map is $K_{a,b}$ with $a < b$. Direct calculation, respecting the left action in (3), gives

$$(6) \quad F_u \circ K_{a,b} = \begin{cases} K_{\max\{a,u\},b}, & u < b, \\ \kappa_u, & u \geq b, \end{cases}$$

$$(7) \quad C_u \circ K_{a,b} = \begin{cases} K_{a,\min\{b,u\}}, & u > a, \\ \kappa_u, & u \leq a. \end{cases}$$

Composing any generator with a constant map again gives a constant map, although its value may change.

As long as no constant has appeared, (6) and (7) show inductively that the current endpoints are exactly (A_k, B_k) . If $A_n < B_n$, then monotonicity of the records gives $A_k \leq A_n < B_n \leq B_k$ for every $k \leq n$, so no earlier collapse was possible and the first alternative follows. If $A_n \geq B_n$, there is a first k at which the inequality fails; the crossing update is one of the constant branches in (6)–(7), and all later maps remain constant. $\square$

For the random model, let $(U_i)_{i \geq 1}$ be iid with an atomless Borel probability law ν on I . Independently, let (ε_i) be iid with

$$(8) \quad \mathbb{P}(\varepsilon_i = F) = p, \quad \mathbb{P}(\varepsilon_i = C) = q := 1 - p,$$

and put $g_i = F_{U_i}$ or C_{U_i} according to ε_i . Define the synchronization time

$$(9) \quad T = \inf\{n \geq 1 : \Phi_n(I) \text{ is a singleton}\},$$

with $\inf \emptyset = \infty$. By [theorem 2.1](#), T is exactly the first threshold-order crossing time $A_n \geq B_n$. Atomlessness implies that sampled thresholds are pairwise distinct almost surely, so equality has probability zero in the stochastic statements below.

3. DISTRIBUTION-FREE SYNCHRONIZATION LAW

Write $N_n = \#\{1 \leq i \leq n : \varepsilon_i = F\}$. Whenever $\mathbb{P}(N_n = j) > 0$, survival conditional on $N_n = j$ means that all j floor thresholds are smaller than all $n - j$ cap thresholds. Exchangeability of the distinct iid ranks makes this one favorable choice among $\binom{n}{j}$ choices of the floor ranks. Here every $0 \leq j \leq n$ is reachable when $0 < p < 1$; at $p = 0$ only $j = 0$ is reachable, and at $p = 1$ only $j = n$ is reachable.

Theorem 3.1 (Exact survival law and geometric decomposition). *For every atomless threshold law ν , every $0 \leq p \leq 1$, and every $n \geq 0$,*

$$(10) \quad s_n(p) := \mathbb{P}(T > n) = \sum_{j=0}^n p^j q^{n-j}.$$

Equivalently,

$$(11) \quad s_n(p) = \begin{cases} \frac{p^{n+1} - q^{n+1}}{p - q}, & p \neq \frac{1}{2}, \\ (n+1)2^{-n}, & p = \frac{1}{2}. \end{cases}$$

If $0 < p < 1$, then $T < \infty$ almost surely and

$$(12) \quad \sum_{n \geq 0} s_n(p) z^n = \frac{1}{(1 - pz)(1 - qz)}, \quad \mathbb{E}z^T = \frac{pqz^2}{(1 - pz)(1 - qz)}$$

for $|z| < 1/\max\{p, q\}$. Consequently,

$$(13) \quad T \stackrel{d}{=} G_p + G_q,$$

where G_p, G_q are independent and $\mathbb{P}(G_r = k) = r(1 - r)^{k-1}$ for $k \geq 1$. In particular,

$$(14) \quad \mathbb{E}T = \frac{1}{pq}, \quad \text{Var}(T) = \frac{1 - 3pq}{p^2 q^2}.$$

Equivalently, for $m \geq 2$,

$$(15) \quad \mathbb{P}(T = m) = pq \sum_{k=0}^{m-2} q^k p^{m-2-k};$$

at $p = q = 1/2$ this is $(m - 1)2^{-m}$. At $p = 0$ or $p = 1$, one instead has $T = \infty$ almost surely and $s_n(p) = 1$ for every n .

Proof. Fix j with $\mathbb{P}(N_n = j) > 0$. Conditional on $N_n = j$, all $\binom{n}{j}$ assignments of the floor labels to the ordered thresholds have equal probability. The event $A_n < B_n$ occurs only when the floors receive the j smallest ranks, and that assignment is unique. Hence

$$\mathbb{P}(T > n \mid N_n = j) = \binom{n}{j}^{-1}.$$

Multiplication by $\mathbb{P}(N_n = j) = \binom{n}{j} p^j q^{n-j}$ and summation over the reachable j proves (10); the omitted terms have zero binomial mass. The finite geometric sum gives (11).

For $0 < p < 1$, (11) tends to zero, proving almost-sure finiteness. Summing (10) first in n and then in j gives the first identity in (12). For a positive integer-valued random variable,

$$\sum_{n \geq 0} \mathbb{P}(T > n) z^n = \frac{1 - \mathbb{E}z^T}{1 - z};$$

substitution yields the second identity. This is the product of the pgfs $pz/(1 - qz)$ and $qz/(1 - pz)$, proving (13). Adding the means and variances of the two geometric variables gives

$$\mathbb{E}T = \frac{1}{p} + \frac{1}{q} = \frac{1}{pq}, \quad \text{Var}(T) = \frac{q}{p^2} + \frac{p}{q^2} = \frac{1 - 3pq}{p^2 q^2}.$$

Convolving the two displayed geometric masses gives (15). At either endpoint all generators have the same type. Atomlessness gives $U_i \in (0, 1)$ almost surely for each fixed i , so every finite pure cap or pure floor composition has a nondegenerate interval image. Thus T is infinite almost surely. $\square$

The critical polynomial factor in (11) is exact. It also marks the crossover between two off-critical geometric tails.

Corollary 3.2 (Critical and off-critical tails). *At $p = 1/2$,*

$$\mathbb{P}(T > n) = (n + 1)2^{-n}.$$

If $p \neq 1/2$ and $r = \max\{p, q\}$, $a = \min\{p, q\}$, then

$$(16) \quad \mathbb{P}(T > n) = \frac{r^{n+1}}{r - a} \left[1 - \left(\frac{a}{r} \right)^{n+1} \right] \sim \frac{r^{n+1}}{|2p - 1|}.$$

4. UNIFORM THRESHOLDS AND IMAGE DIAMETER

The preceding law is distribution-free, but metric image size remembers the threshold distribution. We now specialize only this section to $\nu = \text{Leb}|_{[0,1]}$ and set

$$(17) \quad D_n = \mathbb{E}[\text{diam } \Phi_n(I)].$$

Theorem 4.1 (Exact annealed diameter). *For uniform thresholds, every $n \geq 0$, and every j for which $\mathbb{P}(N_n = j) > 0$,*

$$(18) \quad \mathbb{E}[\text{diam } \Phi_n(I) \mid N_n = j] = \frac{1}{(n + 1) \binom{n}{j}}, \quad 0 \leq j \leq n.$$

For every $0 \leq p \leq 1$, including both endpoints, the unconditional identity is

$$(19) \quad \boxed{D_n = \frac{s_n(p)}{n + 1}}.$$

In particular,

$$(20) \quad D_n = \begin{cases} 2^{-n}, & p = \frac{1}{2}, \\ (n + 1)^{-1}, & p \in \{0, 1\}, \end{cases}$$

Define the signed annealed logarithmic rate $\lambda_a(p) = \lim_{n \rightarrow \infty} n^{-1} \log D_n$ and the positive decay exponent $\gamma_a(p) = -\lambda_a(p)$. Then

$$(21) \quad \lambda_a(p) = \begin{cases} \log \max\{p, 1 - p\}, & 0 < p < 1, \\ 0, & p \in \{0, 1\}. \end{cases} \quad \gamma_a(p) = -\lambda_a(p).$$

For $p \neq 1/2$ in the open interval, with r, a as in [corollary 3.2](#),

$$(22) \quad D_n \sim \frac{r^{n+1}}{(n + 1)(r - a)}.$$

Proof. Let $V_{(1)} < \dots < V_{(n)}$ be the order statistics of the n uniform thresholds, with $V_{(0)} = 0$ and $V_{(n+1)} = 1$. Fix a deterministic full type word containing j floors. Over the random thresholds, survival has probability $\binom{n}{j}^{-1}$. On that event the image is exactly $[V_{(j)}, V_{(j+1)}]$; off that event its diameter is zero. The rank labels are independent of the ordered values. Moreover, the $n + 1$ uniform spacings are exchangeable and sum to one, so each has mean $1/(n + 1)$. Every type word with j floors therefore has the same expected diameter. Whenever $\mathbb{P}(N_n = j) > 0$, averaging over the reachable type words proves (18).

If $0 < p < 1$, the law of total expectation and the binomial law of N_n give (19). The endpoints are checked without conditioning on inaccessible counts. For $n \geq 1$, at $p = 0$ the composition is the pure cap $C_{\min_{1 \leq i \leq n} U_i}$ and its image diameter has mean $1/(n + 1)$. At $p = 1$ the composition is the pure floor $F_{\max_{1 \leq i \leq n} U_i}$ and the complementary maximum has the same

mean. The case $n = 0$ is the identity map. Since $s_n(0) = s_n(1) = 1$, this proves (19) at both endpoints. The special cases now follow from (11), and (21)–(22) follow from corollary 3.2. $\square$

Remark 4.2 (Quenched absorption versus annealed decay). For every $0 < p < 1$, theorem 3.1 says that almost every environment has

$$\text{diam } \Phi_n(I) = 0 \quad \text{for all } n \geq T.$$

Thus the sample path undergoes exact finite-time absorption. If one formally permits $\log 0 = -\infty$, its eventual logarithmic rate is $-\infty$; this is not a conventional finite Lyapunov exponent. By contrast, $D_n > 0$ at every finite n and has the finite annealed rate (21). At $p = 0$ or $p = 1$ under uniform thresholds, the diameter is positive at every finite time almost surely but decreases to zero, with mean $(n+1)^{-1}$. Indeed, in the pure-cap case $\mathbb{P}(\min_{i \leq n} U_i > \epsilon) = (1-\epsilon)^n$, and the pure-floor case is symmetric. These endpoint paths never synchronize in the finite-time sense of (9).

The critical identity $D_n = 2^{-n}$ is worth isolating: the factor $n+1$ in the survival probability is canceled exactly by the mean order-statistic spacing. Hence critical survival has a polynomial correction although the mean image diameter does not.

5. EXACT CONTROLS AND SCOPE BOUNDARY

The accompanying standard-library script performs four independent exact lanes. It exhausts type words and rank permutations through length seven, compares chronological pointwise composition with the normal-form update, and checks the crossing criterion. A separately organized fixed-type enumeration verifies the conditional rank probability and spacing factor. Rational arithmetic checks the survival formulas, convolution law, moments, critical and off-critical identities, and endpoint formulas. The script is a finite control, not a substitute for the proofs above; its frozen output and claim map are included in the evidence package.

The owner boundary is equally important. We do not claim ownership of iterated random functions or their stability theory [1], general synchronization results for monotone random maps [2], contraction semigroups on finite chains [3], or standard exchangeability and order-statistic facts. The paper’s auditable residual contribution is the exact conjunction of theorems 2.1, 3.1 and 4.1 for the specified continuous-interval cap–floor family. Specialist comparison against the closest literature is still required before any external novelty language.

6. CONCLUSION

Random cap and floor maps admit a complete finite normal form. That form turns synchronization into a rank-separation problem and yields a distribution-free survival law. For mixed types, the synchronization time is a sum of two independent geometric variables; for uniform thresholds, the surviving interval contributes one mean spacing and gives the exact annealed diameter. The resulting finite absorption, critical prefactor, and endpoint behavior are elementary but sharply distinct. External circulation remains on hold pending a direct-owner literature review.

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